Characterization of OsRGF1 Gene Function in the Root of *Oryza sativa*

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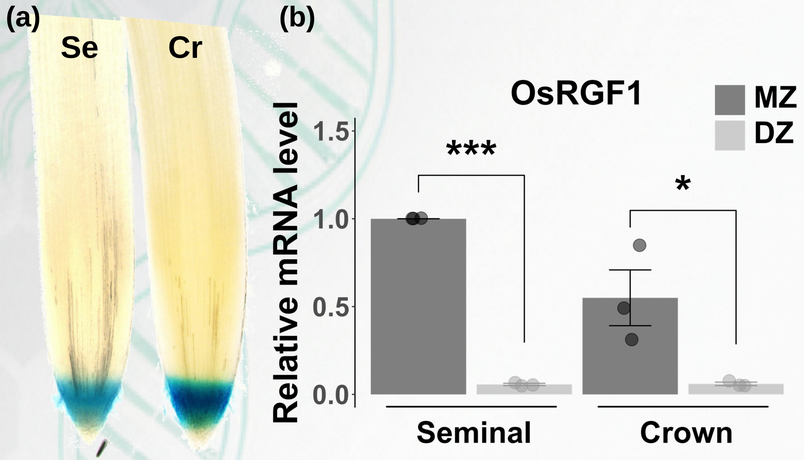


Figure 1. Localization of OsRGF1. Promoter-GUS staining of the seminal (Se) and crown (Cr) root of pOsRGF1::GUS (a). Tissue-specific qPCR analysis of the meristematic zone (MZ) and differentiation zone (DZ) of wild-type root (b).

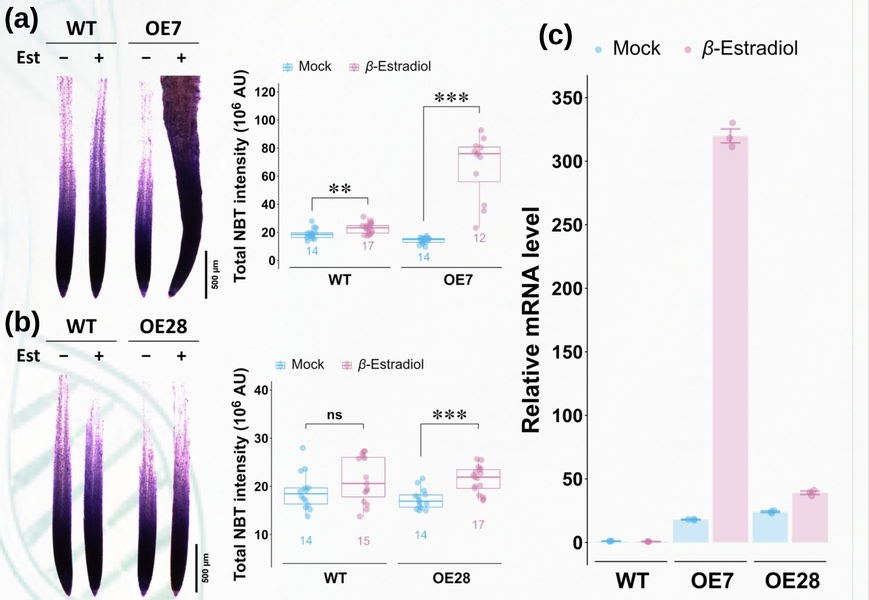


Figure 2. Overexpression of OsRGF1. The total NBT intensity of two independent estradiol-inducible overexpression lines, XVE-OsRGF1-7 (OE7) (a) and XVE-OsRGF1-28 (OE28) (b) after 6 days beta-estradiol treatment. The relative mRNA expression level of OE7 and OE28 (c), PP2AA as the control .

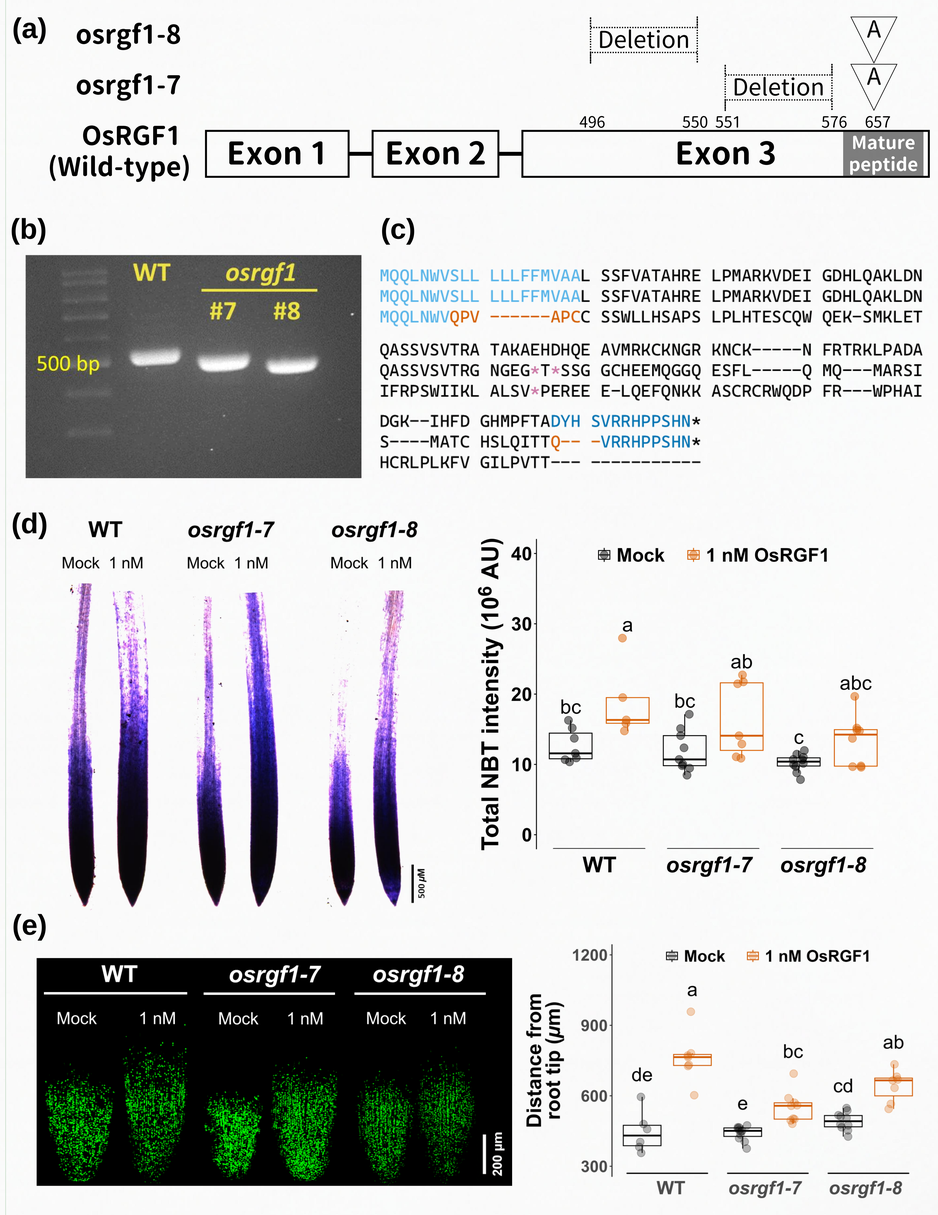


Figure 3. Loss-of-function of OsRGF1. The two independent CRISPR lines, osrgf1-7 and osrgf1-8, exhibit large deletion and single insertion in the OsRGF1 gene (a). The full length cDNA of the OsRGF1 (b). The amino sequence translated from the cDNA Sanger sequencing results (c). The total NBT intensity (d) and total EdU intensity (e) of the mutants.

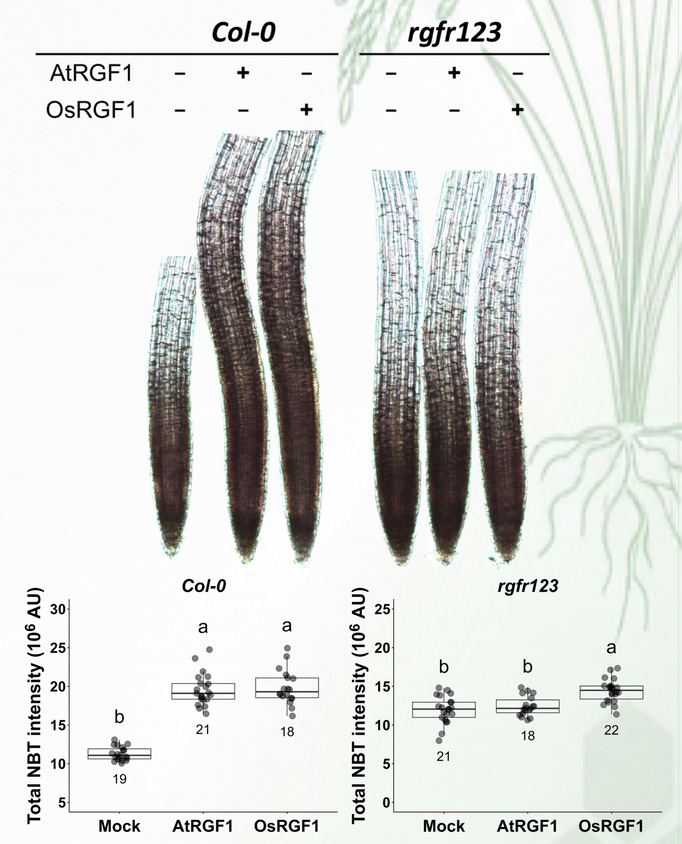


Figure 4. OsRGF1 and AtRGF1 functions similarly in Arabidopsis root.

In progress

- OE lines screening
- Native peptide classification
- Abiotic stress testing

Acknowledgement